ENGR 202 K-map & boolean-eqn

Initialization Fault Report Inputs

Note: V1.00; The output will reset the asynchronous D flip-flop & the output from logic gate will have a counter to then flip the bits to the next state.

1. MCU initialization

* MCU resets flipflop.
  will finish diagnostic-self & change state to 001

2. Power rail stability
   * Before entering main ~~component~~, there will be a counter & if voltage meet requirement then will enter

A = magnet_rail_line
B = voltage ref.
C = ADC
D = High precision Amp
E = temp sensor
F = accelerometer
G = gyro
H = ARINC IC
I = pressure sensor

IC & sensors

* There are more components but this is the general idea
  - we will be using these throughout

next Page

* Due to knowing the boolean eq-n before hand, since it is only 1 output that will work & doing the K-map for 9 vars, would be tedious, so I will be showing work to get only what I will be using for the boolean eq-n

ABCD \ EFGHI	11111
	1

* using NAND & Not gates
* $f_{rail} = \overline{(A \cdot B \cdot C \cdot D \cdot E \cdot F \cdot G \cdot H \cdot I)}$

3. Reset Components
   * some components have a physical-reset pin while others need to be through software. Some sensors don't have one.

- Physical Reset

A1 \ A2	00	10	11	10
00	0	0	0	0
01	0	0	0	0

I4 \ I0	0	0	1	0
10	0	0	0	0

$V_{phys\ reset} = \overline{(A_1 \cdot A_2 \cdot C \cdot H)}$; NAND & NOT Gates

- Software Reset: Note accel. are split & only component that does soft. reset thus will be doing logic for it

F1 \ F2	0	1
0	0	0
1	0	1

$\Rightarrow R_{soft.\ reset} = \overline{(F_1 \cdot F_2)}$

Next Page